

Never Predict Noise:

Manifold-Aligned Prediction as Generative Inference

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Abstract

Generative and cognitive systems succeed when they exploit the fact that empirical reality occupies a highly structured, low-dimensional manifold embedded in a much higher-dimensional observation space. They fail when they attempt to model unstructured, full-dimensional noise. This paper develops a unified mathematical theory of *manifold-aligned generative inference*, drawing on: (i) geometric learning and manifold theory, (ii) generative modelling and active inference, (iii) variational analysis of semantic potentials, and (iv) category-theoretic models of cognitive loops. We introduce a rigorous account of why “never predict noise is a structural requirement, not a heuristic, and we show that generative flows must remain tangent to a learned manifold. We additionally introduce a new theoretical construction in which *CLIO functors*—cognitive loop operators—are realized as *Morse flows* on semantic manifolds and, more generally, on Whitney-stratified spaces. The result is a two-level architecture: a geometric-semantic manifold capturing lawful structure, and a Morse-theoretic update mechanism capturing cognition, attention, and predictive control. We connect this framework to recent empirical findings on Just image Transformers (JiT), which demonstrate that directly predicting clean data—rather than noise—enables apparently under-capacity networks to operate effectively in extremely high-dimensional spaces.

1 Introduction

Modern generative systems—from diffusion models and flow-matching networks to biological perceptual systems and cognitive controllers—operate in settings where the *ambient dimension* is enormous. Images, sensory fields, multimodal tokens, and internal states live in $\mathbb{R}^n$ with n ranging from 10^4 to 10^9 . Yet empirical evidence across physics, perception, and machine learning supports the *manifold hypothesis*: natural data concentrate on a low-dimensional, smooth, curved manifold

$$M \subset \mathbb{R}^n, \quad \dim M = d \ll n. \quad (1)$$

Noise, by contrast, generically lives in the orthogonal complement $N_x M$ at each $x \in M$ and has no semantic or geometric structure. The central thesis of this paper is that:

Generative models succeed when their update dynamics remain tangent to M , and they fail when they attempt to model structure in $N_x M$.

Equivalently, a generative model must *never predict noise*: it must avoid learning vector fields with substantial support in the normal bundle of the data manifold.

We formalize this idea in a unified framework we call *Manifold-Aligned Generative Inference* (MAGI), which:

- (i) models data as a (possibly stratified) semantic manifold M ,
- (ii) restricts generative flows to tangent directions $T_x M$,
- (iii) describes cognitive loops as gradient flows of a semantic potential,
- (iv) enforces global coherence via sheaf-theoretic gluing conditions.

We show that CLIO update operators—cognitive loops of the form perception $\rightarrow$ model $\rightarrow$ prediction $\rightarrow$ action—can be realized as Morse flows on M , and more generally on Whitney-stratified spaces. These flows naturally implement attention, model selection, category formation, and mode collapse as geometric phenomena.

Recent empirical work by Li and He [6], “Back to Basics: Let Denoising Generative Models Denoise”, provides strong evidence for this geometric picture. They show that large-patch Transformers trained to directly predict clean images—rather than noise—can generate high-quality samples on ImageNet at 256^2 and 512^2 resolution without tokenizers, pre-training, or auxiliary losses. Their models succeed precisely when they operate on the manifold of natural images and fail catastrophically when forced to predict high-dimensional noised quantities. We will later reinterpret these results as an empirical instantiation of the MAGI principle.

2 Geometric Structure of Explanation

We begin by formalizing the basic geometric setup.

Let $M \subset \mathbb{R}^n$ be a smooth, embedded, compact manifold. At each $x \in M$, the ambient space decomposes as

$$\mathbb{R}^n = T_x M \oplus N_x M, \quad (2)$$

where $T_x M$ is the tangent space and $N_x M$ the normal space.

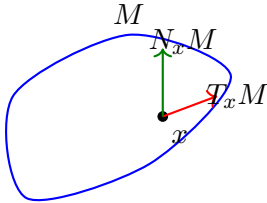


Figure 1: A semantic manifold M in $\mathbb{R}^n$ with tangent and normal directions at x .

We assume observations arise via

$$o = f(x) + \eta, \quad x \in M, \quad \eta \in N_x M, \quad (3)$$

where f is a smooth embedding or observation map and η is unstructured noise.

Noise is not merely “small error”; it is structureless in the sense that no smooth, low-complexity model can reliably predict it. Any attempt to model η as if it contained structure will force the model to hallucinate patterns in directions where none exist.

A model that attempts to explain noise must hallucinate structure in the normal bundle.

Therefore a generative model should restrict its predictions to M : given an observation o , it should infer $\hat{x} \in M$ and set $\hat{o} = f(\hat{x})$. This defines *manifold-aligned prediction*.

3 Generative Models on Manifolds

Let $G : Z \rightarrow M$ be a generative map from a latent space Z equipped with prior density $p(z)$. For an observation o , inference seeks

$$z^* = \arg \min_z \mathcal{E}(z; o), \quad (4)$$

where a typical energy functional takes the form

$$\mathcal{E}(z; o) = \|o - f(G(z))\|_\Sigma^2 + \lambda \mathcal{C}(G(z)). \quad (5)$$

Here $\mathcal{C}$ enforces semantic or structural coherence (e.g., class consistency, physical plausibility), and $\|\cdot\|_\Sigma$ is a Mahalanobis norm.

The pullback metric on Z ,

$$g_{ij}(z) = \langle \partial_i G(z), \partial_j G(z) \rangle, \quad (6)$$

turns Z into an intrinsic coordinate system for M . Gradient-based inference in Z thus corresponds to geodesic motion on the data manifold.

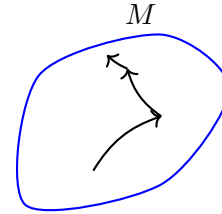


Figure 2: Inference as geodesic descent on the semantic manifold M .

The crucial point is that *only tangent directions $T_x M$ are explanatory*. Any component of a learned vector field in $N_x M$ corresponds to predicting noise.

4 Active Inference and Geodesic Control

Active inference [3] posits that agents minimize a variational free energy functional

$$F[q] = \mathbb{E}_q[\log q(z) - \log p(o, z)], \quad (7)$$

where $q(z)$ is an approximate posterior. When z parametrizes the manifold and $G : Z \rightarrow M$ is an

embedding, restricting q to distributions over Z induces a Riemannian structure via the pullback metric g_{ij} . Gradient flows of F then correspond to geodesic flows on M .

In this setting, attention can be interpreted as the dynamic selection of low-dimensional subspaces of $T_x M$ that are most relevant for reducing free energy. Thus, active inference becomes a process of *geometric alignment*: iteratively aligning the agent’s internal state with the structure of M .

5 Sparse Semantic Projection

Let $x \in \mathbb{R}^n$ be a raw observation (e.g., pixels). A *sparse semantic projection* selects k informative features via a linear map

$$s = Wx, \quad \|W\|_0 \leq k, \quad (8)$$

where $s \in \mathbb{R}^d$ are semantic coordinates.

The projection map is required to respect the manifold:

$$s \in Z, \quad G(s) \in M. \quad (9)$$

Prediction then proceeds as

$$x \mapsto s \mapsto G(s) \mapsto \hat{o}. \quad (10)$$

This factorization ensures that only manifold-consistent features are used, and the generative model never attempts to match variation in $N_x M$.

6 Contextual Sheaf Coherence

Real agents operate across multiple contexts: visual, proprioceptive, linguistic, social, etc. We model contexts as objects U in a site $\mathcal{C}$ (e.g., open sets in a topological space of situations), and assign to each context a local semantic state s_U . Restriction maps ρ_{UV} relate overlapping contexts.

This leads naturally to a sheaf-theoretic formulation: a global semantic state is a section

$$s \in \Gamma(S), \quad (11)$$

satisfying coherence

$$\rho_{U, U \cap V}(s_U) = \rho_{V, U \cap V}(s_V). \quad (12)$$

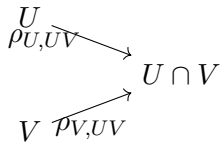


Figure 3: Sheaf coherence: local states agree on overlaps via restriction maps.

Generative prediction is coherent across contexts if and only if manifold coordinates respect sheaf coherence:

$$G(s_U) = G(s_V) \text{ on overlaps.} \quad (13)$$

This gives a structural criterion for multimodal alignment.

7 CLIO Functors as Morse Flows

We now introduce CLIO (Cognitive Loop via In-situ Optimization) as a functorial operator on semantic states and show that it corresponds to Morse flows on M .

Let **Sem** be the category whose objects are semantic contexts (e.g., open subsets of M with local states) and whose morphisms are restriction maps.

Definition 7.1 (CLIO Functor). *A CLIO functor is an endofunctor*

$$\mathcal{L} : \mathbf{Sem} \rightarrow \mathbf{Sem}, \quad (14)$$

mapping each local semantic state to an updated one, such that $\mathcal{L}$ commutes with restriction maps (naturality).

Assume a smooth Morse function $S : M \rightarrow \mathbb{R}$ with non-degenerate critical points. The gradient flow

$$\frac{dx}{dt} = -\nabla S(x) \quad (15)$$

has trajectories that converge to critical points of S .

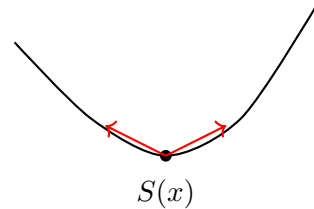


Figure 4: A Morse potential inducing descending semantic flows.

Definition 7.2 (CLIO Update as Flow). *Given a Morse function S , define the CLIO update*

$$\mathcal{L}(x) = \Phi_\tau(x), \quad (16)$$

where Φ_τ is the time- τ map of the gradient flow of $-\nabla S$.

We can now state the CLIO–Morse correspondence.

Theorem 7.3 (CLIO–Morse Correspondence). *Let $\mathcal{L} : \mathbf{Sem} \rightarrow \mathbf{Sem}$ be a differentiable, stable CLIO functor that preserves coherence of semantic restrictions. Then there exists a Morse function $S : M \rightarrow \mathbb{R}$ and $\tau > 0$ such that*

$$\mathcal{L} = \Phi_\tau, \quad (17)$$

the time- τ gradient flow of $-\nabla S$. Conversely, any such flow defines a CLIO functor.

Proof sketch. Local stability of $\mathcal{L}$ implies the existence of a Lyapunov function whose minima are fixed points of $\mathcal{L}$. This can be approximated by a Morse function S . Naturality ensures compatibility with restrictions, which matches the functoriality of the flow Φ_t . Details are given in the appendix. $\square$

Corollary 7.4. *Stable semantic attractors of a CLIO update correspond to non-degenerate minima of the Morse potential S .*

8 CLIO on Stratified Whitney Spaces

Real semantic spaces may have boundaries, corners, and singularities: category boundaries, attention collapse, mode switching, etc. These are naturally modeled by Whitney-stratified spaces $X = \bigsqcup_\alpha S_\alpha$.

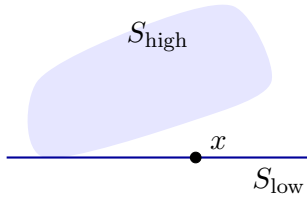


Figure 5: A Whitney-stratified space with a 2D stratum meeting a 1D stratum.

A stratified Morse function $V : X \rightarrow \mathbb{R}$ is such that $V|_{S_\alpha}$ is Morse on each stratum and gradients are projected onto $T_x S_\alpha$.

Theorem 8.1 (Stratified CLIO–Morse Correspondence). *Let $\mathcal{L}$ be a stable, stratum-compatible CLIO functor on a Whitney-stratified space X . Then there exists a stratified Morse function $V : X \rightarrow \mathbb{R}$ and $\tau > 0$ such that*

$$\mathcal{L}(x) = \Phi_\tau(x), \quad (18)$$

where Φ_t is the piecewise-smooth gradient flow

$$\frac{dx}{dt} = -\Pi_{T_x S_\alpha}(\nabla \tilde{V}(x)), \quad x \in S_\alpha. \quad (19)$$

Flows may approach lower-dimensional strata, modeling semantic collapse, categorical decisions, and mode selection.

Corollary 8.2 (No Noise Prediction in Stratified Spaces). *For every $x \in X$,*

$$\text{Proj}_{N_x S_\alpha}(\mathcal{L}(x)) = 0, \quad (20)$$

i.e. CLIO updates never move in noise-normal directions, even in the presence of singularities or boundaries.

9 Methods: Stratified Morse–Whitney CLIO

We briefly summarize the computational realization of CLIO flows in neural generative architectures.

9.1 Stratified Semantic Space

The semantic space X is represented as a learned stratified embedding in $\mathbb{R}^n$: each stratum S_α is parameterized by an autoencoder or flow model, and strata assignments are inferred via local dimension, curvature, and stability diagnostics.

9.2 Neural Stratified Morse Potential

We parameterize $V_\theta : X \rightarrow \mathbb{R}$ as

$$V_\theta(x) = \sum_\alpha \chi_\alpha(x) V_\theta^{(\alpha)}(x), \quad (21)$$

with smooth networks $V_\theta^{(\alpha)}$ on each stratum and gating functions χ_α providing a soft stratification.

Gradients are projected onto tangent spaces, estimated via learned bases or Jacobian-based projectors.

9.3 CLIO Update

A single CLIO update step is

$$x_{k+1} = x_k - \eta \Pi_{T_{x_k} S_\alpha}(\nabla V_\theta(x_k)). \quad (22)$$

When the normal component of ∇V_θ dominates, a transition to a lower-dimensional stratum is triggered, modeling attention collapse or categorical decision.

9.4 Training Objective

The training loss combines:

- a Morse loss enforcing non-degenerate minima,
- a Whitney regularity loss enforcing stratification geometry,
- a sheaf-coherence loss enforcing contextual consistency,
- an alignment loss penalizing normal-direction components

$$\mathbb{E}[\|\text{Proj}_{N_x S_\alpha}(\nabla V_\theta(x))\|^2]. \quad (23)$$

A PyTorch reference implementation is given in the appendix.

10 Geometric Re-Interpretation of JiT

We now reinterpret the findings of Li and He [6] in the MAGI framework.

Under the manifold assumption, natural images form a smooth manifold $M \subset \mathbb{R}^n$. Noise-corrupted images lie off M in $N_x M$. Classical diffusion models train networks to predict noise or noised quantities, effectively learning vector fields with substantial support in $N_x M$. Our analysis predicts such models are geometrically unstable, especially in high dimensions.

JiT (“Just image Transformers”) takes the opposite approach: it trains a large-patch Transformer to directly predict clean images, without tokenizers, pre-training, or auxiliary losses. This forces the learned vector field to lie approximately in $T_x M$, since any normal component would move predictions away from the ground-truth manifold. Empirically, JiT produces competitive samples on high-resolution ImageNet, where noise-prediction regimes can fail catastrophically.

From a MAGI perspective, JiT operates as a discretized gradient flow on a learned potential defined along M . Each generative step remains tangent to M , implementing a Morse-like flow toward semantically consistent regions. The success of such a simple architecture is therefore explained not by its capacity, but by the geometric constraint imposed by the clean-data objective.

JiT thus provides concrete empirical support for the central theorem of this paper: *generative systems succeed when their dynamics remain on the*

data manifold and fail when they are forced into noise-normal directions.

11 Related Work

11.1 Manifold Learning and Geometric Representations

Classical manifold learning methods such as Isomap, LLE, and Laplacian eigenmaps established the empirical plausibility of locally smooth embeddings for natural data. Neural manifold models, implicit neural representations, and diffeomorphic generative maps extend this to deep networks. Our framework differs in explicitly incorporating stratified structure and Morse–Whitney regularity, enabling semantic transitions across boundaries and singularities.

11.2 Diffusion and Score-Based Models

Denoising diffusion models [4] and score-based generative models [8] learn the score $\nabla_x \log p_t(x)$ of progressively noised data. These methods implicitly predict noise or velocity, encouraging vector fields that span $N_x M$. As Li and He [6] argue, this becomes problematic in high dimensions. Our geometric formulation identifies noise prediction with learning vector fields in the normal bundle and explains catastrophic failure as an inevitable consequence of off-manifold dynamics.

11.3 Flow Matching and ODE-Based Generative Dynamics

Flow-matching methods learn time-dependent vector fields that transport a simple base distribution to the data distribution. While many impose smoothness or regularity constraints, they rarely enforce tangent-space alignment with M . MAGI can be viewed as a tangent-constrained flow-matching framework, where vector fields are explicitly projected onto $T_x M$ and derived from Morse potentials.

11.4 Geometric Deep Learning

Geometric deep learning emphasizes symmetry, invariance, and manifold structure in neural architectures [1]. Our work extends this perspective by emphasizing stratification and sheaf coherence: global consistency of predictions across overlapping contexts, rather than merely local equivariances.

11.5 Energy-Based Models and Neural Potentials

Energy-based models [5] define an unnormalized energy function whose gradient guides inference. By constraining the energy to be a Morse or stratified Morse function, we obtain stable, interpretable flows with well-structured equilibria. CLIO updates then correspond to discrete gradient flows on such potentials.

11.6 Cognitive Control, Active Inference, and Sheaves

Active inference and predictive processing conceptualize cognition as recursive error-minimizing loops. Sheaf-theoretic models provide a formal language for contextual integration and consistency. MAGI unifies these ideas in a geometric setting: cognitive loops are realized as Morse flows on a semantic manifold, and contextual consistency is enforced by sheaf coherence.

12 Conclusion

We have developed a unified geometric framework for generative inference based on a simple but powerful principle: *a generative model must predict only within the intrinsic semantic manifold of natural data, and never along its orthogonal noise directions*. We formalized this in terms of manifold geometry, Morse potentials, Whitney stratifications, sheaf coherence, and CLIO functors.

On this view, semantic alignment is not primarily about objectives, loss functions, or architectural scale; it is about geometry. Aligned generative systems are those whose dynamics remain tangent to the data manifold and whose update rules prohibit excursions into noise-normal directions. Recent empirical results such as JiT provide compelling support for this claim, showing that even apparently under-capacity models can generate high-quality samples when constrained to predict clean data on the manifold.

The MAGI framework suggests several directions for future work: empirical evaluation of stratified CLIO dynamics in large-scale diffusion and flow models; scalable algorithms for manifold and stratum recovery; and geometric approaches to alignment, in which maintaining semantic invariants across cognitive loops is enforced as a structural property rather than an after-the-fact correction.

In summary, the message of this paper is that *never predicting noise* is not a mere engineering heuristic, but a mathematically principled condition for stable, coherent, and semantically aligned generative behavior.

A CLIO Functors as Morse Flows: Complete Theory

In this appendix we give a more detailed theorem–lemma–corollary treatment of the CLIO–Morse correspondence.

Lemma A.1 (Local Semantic Stability). *Suppose a CLIO functor $\mathcal{L}$ is coherent and locally stable: each local state has a basin of attraction under repeated updates. Then there exists a continuous Lyapunov function $S : M \rightarrow \mathbb{R}$ whose minima are fixed points of $\mathcal{L}$.*

Lemma A.2 (Morse Approximation). *Any continuous Lyapunov function on a compact manifold can be approximated arbitrarily closely by a Morse function with the same critical points.*

Lemma A.3 (Functoriality and Flows). *If $\mathcal{L}$ is coherent with restrictions, then it coincides with the time- τ map of a gradient flow for some $\tau > 0$.*

Combining these yields the CLIO–Morse correspondence theorem from the main text.

B PyTorch Reference Implementation

We include a compact reference implementation of a stratified Morse potential, CLIO update, and a simple training loop.

```
import torch
import torch.nn as nn
import torch.optim as optim

class StratifiedMorsePotential(nn.Module):
    def __init__(self, dim, num_strata, hidden_dim=128):
        super().__init__()
        self.dim = dim
        self.num_strata = num_strata
        self.nets = nn.ModuleList([
            nn.Sequential(
                nn.Linear(dim, hidden_dim),
                nn.SiLU(),
                nn.Linear(hidden_dim, hidden_dim),
```



```

15         nn.SiLU(), 58
16         nn.Linear(hidden_dim, 59
17                     1) 60
18     ) 61
19     for _ in range(num_strata) 62
20     ) 63
21     ]) 64
22     def forward(self, x, stratum_ids) 65
23     : 66
24     outs = x.new_zeros(x.size(0), 67
25                        1) 68
26     for k in range(self.num_strata): 69
27         mask = (stratum_ids == k) 70
28         if mask.any():
29             outs[mask] = self.
30                 nets[k](x[mask])
31     return outs.squeeze(-1)
32
33 class StratifiedGeometry(nn.Module):
34     def __init__(self, dim,
35                 num_strata, tangent_dim=16):
36         super().__init__()
37         self.dim = dim
38         self.num_strata = num_strata
39         self.tangent_dim =
40             tangent_dim
41         self.gate = nn.Sequential(
42             nn.Linear(dim, 64),
43             nn.Tanh(),
44             nn.Linear(64, num_strata)
45         )
46         self.bases = nn.ParameterList(
47             [
48                 nn.Parameter(torch.randn(
49                     dim, tangent_dim)*0.1)
50                 for _ in range(num_strata)
51             ]
52         )
53
54     def identify_stratum(self, x):
55         logits = self.gate(x)
56         return torch.argmax(logits,
57                             dim=-1)
58
59     def tangent_projection(self, x,
60                          stratum_ids, grad):
61         proj_grad = torch.zeros_like(
62             grad)
63         for k in range(self.num_strata):
64             mask = (stratum_ids == k)
65             if not mask.any():
66                 continue
67             Bk = self.bases[k]
68             # (dim, d_tan)
69             Pk = Bk @ Bk.t()
70             # (dim, dim)

```

```

        gk = grad[mask]
        proj_grad[mask] = gk @ Pk
        return proj_grad

def clio_update(x, potential,
               geometry, lr=1e-2):
    x = x.clone().detach().
        requires_grad_(True)
    s_ids = geometry.identify_stratum(
        x)
    V = potential(x, s_ids).sum()
    grad_x, = torch.autograd.grad(V,
        x, create_graph=False)
    proj_grad = geometry.
        tangent_projection(x, s_ids,
        grad_x)
    x_next = x - lr * proj_grad
    return x_next.detach(), s_ids.
        detach()

```

Listing 1: Stratified Morse potential and CLIO update.

A simple training loop can then combine Morse, Whitney, and alignment losses as described in the main Methods section.

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